Perfect-Power Obstructions and Sparse Powerful Values in Cyclic Decimal Concatenations

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Abstract

We prove local obstructions and a quantitative density-zero bound for powerful values among cyclic concatenations of initial positive integers. For decimal digit lengths with $v_3(d) = 1$, a cubic reduction gives an exact local classification: 131 classes of m modulo 243 exclude all rotations over $\mathbb{Q}_3$; every other class has a local square, cube, or fifth power among the first four cuts. When $v_3(d) = 2$, exactly 133 classes modulo 243 exclude all rotations; every other class has a local power at one of seven fixed cuts, all at most 22. When $v_3(d) = 3$, exactly 3673 classes modulo 6561 exclude all rotations, and the first two cuts suffice for every other class. At complete digit boundaries, exact valuations and characters exclude a set of digit lengths of lower density exceeding 95.5995%. A general lifting theorem gives local power equivalences and a quadratic reduction on entire digit intervals. Along suitable progressions of digit lengths, the proportion of m for which no rotation is powerful can be arbitrarily close to one. The quantitative sparsity bound holds in every fixed integer base.

1 Statements

In base $g \geq 2$, let W_m concatenate $1, 2, \ldots, m$, and let $D(m)$ be its digit length, with $W_0 = D(0) = 0$. For $0 \leq k < m$, moving the first k whole integer blocks to the end gives

$$R_{m,k} = g^{D(k)} W_m - (g^{D(m)} - 1) W_k. \tag{1}$$

In base ten these numbers, in increasing order, form OEIS A001292 [1]. Kashihara conjectured that none except 1 is a perfect power. Ripà [2] excludes $m \equiv 2, 3, 5, 6 \pmod 9$ by a digit-sum argument. The full conjecture remains open. We give a uniform sparsity bound and local obstructions for complete digit boundaries.

Recall that a positive integer is *powerful* if every prime divisor has exponent at least two. Every nontrivial perfect power is powerful. Logarithms in the following theorem and in Section 5 are natural.

Theorem 1 (Quantitative sparsity)**.** *For every fixed integer base $g \geq 2$,*

$$\#\{(m,k) : 2 \leq m \leq X,\ 0 \leq k < m,\ R_{m,k} \text{ powerful}\} \ll_g \frac{X^2 (\log\log X)^2}{\log X}. \tag{2}$$

For A001292, the corresponding count among the first N terms is $O(N(\log\log N)^2/\log N)$.

For the next result the base is ten and v_3 denotes the 3-adic valuation.

Theorem 2 (Complete digit boundaries)**.** *Let $d \geq 1$, $m = 10^d - 1$, $h = v_3(d)$, and $u = d/3^h$. Every rotation satisfies*

$$v_3(R_{m,k}) = h + 2, \qquad \frac{R_{m,k}}{3^{h+2}} \equiv u(2 + 6d) \pmod 9. \tag{3}$$

Corollary 1. *Put $v = h + 2$ and let $\mathcal{P}(v)$ be its set of prime divisors. No rotation at $m = 10^d - 1$ is a perfect power in any of the following cases:*

$\mathcal{P}(v)$	*condition on* u
$\{2\}$	$u \equiv 1 \pmod 3$
$\{3\}$	$u \equiv 1, 2, 7, 8 \pmod 9$
$\{2, 3\}$	$u \equiv 1, 7 \pmod 9$.

This includes $d \equiv 3, 6, 9, 21, 24 \pmod{27}$ and $d = 81$.

Proof. A prime exponent of a perfect power must divide v. A square requires v even and its normalized unit congruent to 1 modulo 3; by (3) this requires $u \equiv 2 \pmod 3$. A cube requires $3 \mid v$, hence $h \geq 1$ and $3 \mid d$. Its normalized unit must be $\pm 1 \pmod 9$, so $2u \equiv \pm 1 \pmod 9$, or $u \equiv 4, 5 \pmod 9$. The table excludes every available prime exponent. □

For the additional boundary primes, put

$$(p, s) \in \{(37, 3), (333667, 9)\}, \qquad C_s = \frac{9(11 - 10s)}{200} \pmod p.$$

Define

$$A_0 = 0, \qquad A_r = -9 \sum_{j=1}^{r} \frac{10^{j-1}}{10^j - 1} \pmod p \quad (0 < r < s).$$

All denominators here are units modulo p; $C_3 = 33$ and $C_9 = 221885$.

Theorem 3 (Two affine residue formulas)**.** *For every $d \geq 1$, write $d = sn + r$, $0 \leq r < s$. Then*

$$V_d := W_{10^d - 1} \equiv C_s n + A_r \pmod p, \qquad R_{10^d - 1, k} \equiv 10^{D(k)} V_d \pmod p. \tag{4}$$

Theorem 4 (General boundary-prime lifting)**.** *Let $p \geq 7$ be prime, $r = \operatorname{ord}_p(10)$, and choose P with*

$$r \mid P, \qquad 10^P \equiv 1 \pmod{9r}, \qquad p \nmid P, \qquad C := V_P \not\equiv 0 \pmod p.$$

For every positive d divisible by P, set $a = v_p(d)$. Every whole-block rotation satisfies

$$v_p(R_{10^d - 1, k}) = a, \qquad \frac{R_{10^d - 1, k}}{p^a} \equiv 10^{D(k)} C \frac{d}{P p^a} \pmod p. \tag{5}$$

In particular, $d = Ppt$ with $p \nmid t$ excludes all perfect powers. More generally, every possible prime exponent divides both $v_3(d) + 2$ and $v_p(d)$, for each applicable pair (p, P). The six primes in the table of Section 4 all satisfy the hypotheses. For the two affine formulas take $P = s$ and $C = C_s$. If a prime exponent ℓ divides $(p - 1)/r$, then $Cd/(Pp^a)$ must also be an ℓth power modulo p. In particular, $d = 111t$ with $37 \nmid t$ and $d = 3003003t$ with $333667 \nmid t$ are excluded. The new choice $p = 757, P = 27$ also excludes $d = 327024 = 27 \cdot 757 \cdot 16$, since the valuations of every rotation at 3 and 757 are respectively 5 and 1.

Theorem 5 (Reduction on an entire digit interval)**.** *Let $p \geq 7$ be prime, $r = \operatorname{ord}_p(10)$, and $r \mid P$, $10^P \equiv 1 \pmod{9r}$, $p \nmid P$. Put $C = V_P \pmod p$, allowing $C = 0$. For $P \mid d$, $a = v_p(d)$, every $10^{d-1} \leq m < 10^d$ and $0 \leq k < m$ satisfy*

$$R_{m,k} \equiv 10^{D(k)} \left(\frac{m(m+1)}{2} + C \frac{d}{P} \right) \pmod{p^{a+1}}. \tag{6}$$

If $Pp \mid d$, exactly $2(p - 1)$ residue classes $m \pmod{p^2}$ give $v_p(R_{m,k}) = 1$ for every cut k.

For example, if $d = 111u$, all rotations have valuation one at 37 whenever

$$m \equiv 0, -1 \pmod{37}, \qquad m \not\equiv 296u,\ 1073u - 1 \pmod{1369}.$$

These are 72 residue classes modulo 1369 in every such digit interval.

Corollary 2 (Arbitrarily high coverage of whole rotation blocks)**.** *For every $\varepsilon > 0$ there are positive integers L, d_0 such that, for every $d \geq d_0$ divisible by L, at least $(1 - \varepsilon)9 \cdot 10^{d-1}$ choices of d-digit m have no powerful integer among their rotations.*

Corollary 3 (Density of excluded complete boundaries)**.** *The set of positive integers d for which every rotation at $m = 10^d - 1$ is not a nontrivial perfect power has lower natural density at least*

$$\rho_* = 0.9559952715065166\ldots > 0.95599527150.$$

Theorem 6 (Further complete-boundary exclusions)**.** *No rotation at $m = 10^d - 1$ is a nontrivial perfect power when $1 \leq d \leq 27$, or when $d = 54$ or 243. The same conclusion holds for*

$$\begin{aligned} &d \equiv 12, 15 \pmod{333}, \\ &d \equiv 27 \pmod{37249072371}, \qquad d \equiv 243 \pmod{25475666076}. \end{aligned} \tag{7}$$

The finite inputs to this theorem are exact modular certificates; Section 4 gives their verification and the unbounded period argument.

2 Exact 3-adic values at complete boundaries

Write $V_d = W_{10^d-1}$ and $S(d) = D(10^d - 1)$. We will prove the stronger intermediate congruence

$$\frac{V_d}{9d} \in \mathbb{Z}_3, \qquad \frac{V_d}{9d} \equiv 2 + 6d \pmod 9. \tag{8}$$

Quotients by d in this section are taken in $\mathbb{Q}_3$.

Lemma 1 (Averaging power series)**.** *For integers $d \geq 1$ and $n \geq 1$,*

$$v_3\left(\frac{1}{d}\sum_{j=1}^{d} j^n\right) \geq -\lfloor \log_3(n+1) \rfloor. \tag{9}$$

Suppose $f(X) = \sum_{n\geq 0} c_n X^n$ has $v_3(c_n) \geq n$, $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. Then

$$\frac{1}{d}\sum_{j=1}^{d} f(j) \equiv 2 + 6d \pmod 9. \tag{10}$$

Proof. Using Stirling numbers of the second kind gives the exact identity

$$\frac{1}{d}\sum_{j=1}^{d} j^n = \sum_{r=0}^{n} \left\{ {n+1 \atop r+1} \right\} \frac{r!}{r+1}\binom{d-1}{r}.$$

Each denominator loses at most $\lfloor \log_3(n+1) \rfloor$ factors of 3, proving (9). For $n \geq 3$, $n - \lfloor \log_3(n+1) \rfloor \geq 2$, so all these terms vanish modulo 9 after averaging. The same holds for $n = 2$ by its stronger coefficient bound. Only $c_0 + c_1(d+1)/2$ remains, congruent to $5 + 3(d+1)/2 = 2 + 6d \pmod 9$. The coefficient bounds also justify convergence and termwise averaging. □

Define the 3-adic logarithm explicitly by

$$\ell = \sum_{r\geq 1} \frac{(-1)^{r+1} 9^r}{r}, \qquad v_3(\ell) = 2.$$

In the following power series put

$$E = e^{\ell X}, \quad a = E/10, \quad q = 9a, \quad b = E - 1, \quad S(X) = \frac{(9X-1)E + 1}{9},$$

and define

$$\frac{U(X)}{9} = a^2 + \frac{ab+1}{9}\sum_{r\geq 0}\binom{q}{r+2} b^r, \qquad f(X) = \frac{U(X)}{9} e^{-\ell S(X)}. \tag{11}$$

At every positive integer j, $S(j)$ is the digit length of V_j, and $U(j)$ concatenates all j-digit integers. Indeed, with $a = 10^{j-1}$, $b = 10^j - 1$, and $q = 9a$, the usual arithmetic-geometric sum gives

$$b^2 U(j) = (ab+1)(1+b)^q - ((1+b)b + 1).$$

Expanding $(1+b)^q$ cancels its constant and linear terms and gives (11), since $a + q = 10^j$ and $aq = 9a^2$.

Lemma 2. *The coefficients of f satisfy all hypotheses of Lemma 1.*

Proof. Let $\mathcal{A}$ consist of series whose degree-n coefficient has valuation at least n. It is closed under addition and multiplication. In the rth summand of $U/9$, the factor b^r makes all degrees below r zero. The factor q in $\binom{q}{r+2}$ cancels the displayed denominator 9. The remaining expression is a polynomial in E with 3-adic denominators bounded by $(r+2)!$. Its degree-n coefficient, for $r \leq n$, therefore has valuation at least

$$2n - v_3(n!) - v_3((r+2)!) \geq n.$$

Here $v_3(n!) \leq (n-1)/2$ for $n \geq 1$, and $v_3((r+2)!) \leq (n+1)/2$; the degree-zero case is immediate. The term a^2 satisfies the same bound. Thus $U/9 \in \mathcal{A}$.

The constant term of S is zero. Its linear coefficient is $1-\ell/9$, of valuation at least 2, and for $n \geq 2$ its coefficient is $\ell^{n-1}(n-\ell/9)/n!$, of valuation at least n. Hence $S \in \mathcal{A}$. Expanding $e^{-\ell S}$ preserves these bounds: its rth exponential term gains $2r - v_3(r!) \geq 0$ factors of 3. Consequently $f \in \mathcal{A}$.

Expansion through degree two in (11) gives

$$\begin{aligned} c_0 &= \frac{1}{200}, \\ c_1 &= \frac{169\ell}{3000} + \frac{\ell^2}{1800}, \\ c_2 &= \frac{671\ell^2}{80000} + \frac{353\ell^3}{54000} + \frac{\ell^4}{32400}. \end{aligned}$$

Since $\ell/9 \equiv 1 \pmod 9$, these imply $c_0 \equiv 5 \pmod 9$, $c_1 \equiv 3 \pmod 9$, and $v_3(c_2) \geq 3$. □

Proof of Theorem 2. The concatenation of the complete digit blocks yields

$$V_d = 10^{S(d)} \sum_{j=1}^{d} U(j) 10^{-S(j)} = 9\, 10^{S(d)} \sum_{j=1}^{d} f(j).$$

Lemma 1 and $10^{S(d)} \equiv 1 \pmod 9$ prove (8). Its right side is a 3-adic unit, so $v_3(V_d) = h+2$.

To transfer the normalized unit to every rotation, note that

$$S(d) = \frac{9d(d+1)}{2} \left(1 + \sum_{r=1}^{d-1} \frac{2 \cdot 9^r}{r+2} \binom{d-1}{r} \right). \tag{12}$$

This follows by expanding $S(d) = 9 \sum_{j=1}^{d} j 10^{j-1}$ in powers of 9. Each term after 1 in the bracket is divisible by 3; hence $v_3(S(d)) = 2 + h + v_3(d+1)$. Lifting the exponent gives

$$v_3(10^{S(d)} - 1) = 4 + h + v_3(d+1).$$

In (1), $10^{D(k)} \equiv 1 \pmod 9$ and the second term is divisible by 3^{h+4}. Thus $R_{m,k} \equiv V_d \pmod{3^{h+4}}$. Combining this with (8) proves both assertions in (3). □

A cubic reduction on digit intervals

Write $T_x = x(x+1)/2$.

Theorem 7 (Uniform cubic congruence)**.** *Let m have d decimal digits and $h = v_3(d) \geq 1$. For every whole-block cut,*

$$10^{-D(k)} R_{m,k} \equiv T_m + \frac{3d}{2}(m^3 - m + 9) - 9dmT_k \pmod{3^{h+4}}. \tag{13}$$

In particular, if $3^{h+2} \mid m$, all rotations satisfy

$$R_{m,k} \equiv \frac{m + 27d}{2} \pmod{3^{h+4}}. \tag{14}$$

Proof. Put $B = 10^d$, $\delta = B - 1$, and $t = B - 1 - m$. The exact tail identity in the proof of Theorem 5, expanded through first order in δ, gives

$$W_m \equiv V_d + \frac{t(t-1)}{2} - \delta \frac{t(t^2+5)}{6} \pmod{3^{2h+4}}.$$

Here the first derivative of the tail at $B = 1$ is $t(t+1)(5-2t)/6$; combining it with the factor B^{-t} gives the displayed integer coefficient. The omitted term δV_d also has valuation at least $2h+4$. Substitute $t = \delta - m$. Any loss from the denominator 6 leaves the discarded quadratic terms of valuation at least $2h+3 \geq h+4$. The binomial expansion gives $\delta \equiv 9d \pmod{3^{h+4}}$, while Theorem 2 gives $V_d \equiv 18d \pmod{3^{h+4}}$. Consequently

$$W_m \equiv T_m + \frac{3d}{2}(m^3 - m + 9) \pmod{3^{h+4}}.$$

Also $D(m) = d(m+1) - \delta/9 \equiv dm \pmod{3^{h+2}}$, so $10^{D(m)} - 1 \equiv 9dm \pmod{3^{h+4}}$. Finally $10^{-D(k)} W_k \equiv T_k \pmod 9$ by digit sums. The rotation identity proves (13). If $3^{h+2} \mid m$, the cubic, quadratic, and cut-dependent terms vanish at the stated precision except $m/2 + 27d/2$. Multiplication by $10^{D(k)} \equiv 1 \pmod 9$ preserves this residue, proving (14). □

Corollary 4 (Complete local classification when $v_3(d) = 1$)**.** *Let $v_3(d) = 1$ and $u = d/3$. All rotations of a d-digit m fail to be prime-exponent powers over $\mathbb{Q}_3$ exactly when at least one of the following holds:*

$$m \bmod 9 \in \{2,3,5,6\}, \qquad m \bmod 27 \in \{9,17\}, \qquad m \bmod 243 \in C_u,$$

where

$$C_u = \begin{cases} \{0,27,54,135,189\}, & u \equiv 1 \pmod 3, \\ \{54,108,162,189,216\}, & u \equiv 2 \pmod 3. \end{cases}$$

These are exactly 131 residue classes modulo 243, and every rotation in these classes is excluded from integer perfect powers. In every other class, some $k \in \{0,1,2,3\}$ and $e \in \{2,3,5\}$ satisfy $R_{m,k} \in (\mathbb{Q}_3^\times)^e$.

Proof. A 3-adic unit is a square exactly when it is 1 modulo 3, a cube exactly when it is ± 1 modulo 9, and always a fifth power. The square and fifth-power assertions follow from Hensel's lemma. For cubes, $(1+3x)^3 = 1+9(x+3x^2+3x^3)$, whose expression in parentheses has derivative 1 modulo 3, proves sufficiency; necessity is immediate. In each case the valuation must also be divisible by the exponent. The factor $10^{D(k)}$ preserves these three power criteria.

Modulo 27, (13) reduces to T_m. The first two listed sets have valuation one, or valuation two with nonsquare unit. Outside them, $m \equiv 1 \pmod 3$ gives a fifth-power unit, and $m \equiv 8, 18 \pmod{27}$ gives a square. It remains to consider $m \equiv 0, -1 \pmod{27}$.

For $m \equiv 0 \pmod{27}$, (14) becomes $R_{m,k}/27 \equiv (m/27 + 3u)/2 \pmod 9$. The nine classes of $m/27$ give precisely the five exclusions C_u; every other nonzero residue gives a square or cube. If this residue is zero, $m \equiv -81u \pmod{243}$. The binomial expansion gives $v_3((10^d-1)/(9d)-1) = 2$, so $v_3(D(m)) = 3$ and $v_3(10^{D(m)}-1) = 5$. Since $R_{m,1} = 10R_{m,0} - (10^{D(m)}-1)$, either $R_{m,0}$ or $R_{m,1}$ has valuation exactly 5 and is a fifth power in $\mathbb{Q}_3$.

For $m = 27z - 1$, (13) gives $R_{m,k}/27 \equiv (3u - z)/2 + uT_k \pmod 9$. At $k = 0, 2, 3$, these residues form an entire coset modulo 3 in $\mathbb{Z}/9\mathbb{Z}$. That coset contains 1, 3, or 8, giving a cube or a square. All these cuts exist because m has at least three digits. Finally the three exclusion sets are disjoint and contain $108 + 18 + 5 = 131$ classes modulo 243. □

More generally, (14) shows that if $h+3$ is a power of 2, $d/3^h \equiv 1 \pmod 3$, and $3^{h+4} \mid m$, then every rotation is excluded: its valuation is $h+3$ and its normalized unit is 2 modulo 3.

A finite sufficient test at every higher valuation

For $P \geq 2$, call a nonzero residue $a \pmod{3^P}$ *certifiably excluded* if, writing $b = v_3(a) < P$ and $w = a/3^b$, all the following hold:

1. $b > 0$, and every prime divisor of b belongs to $\{2,3\}$;
2. if $2 \mid b$, then $w \equiv 2 \pmod 3$;
3. if $3 \mid b$, then $P - b \geq 2$ and $w \not\equiv \pm 1 \pmod 9$.

The zero residue is never certifiably excluded. The conditions depend only on the residue, not its representative.

Proposition 1 (A sufficient nine-residue test)**.** *Let m have d decimal digits, $h = v_3(d) \geq 2$, and $P = h + 4$. Set*

$$A_j = T_m + \frac{3d}{2}(m^3 - m + 9) - 9dmT_j \pmod{3^P} \qquad (0 \leq j \leq 8).$$

If all nine A_j are certifiably excluded, every whole-block rotation of $12\cdots m$ fails to be a prime-exponent power over $\mathbb{Q}_3$, and consequently fails to be an integer perfect power. This is a sufficient test, not an asserted equivalence.

Proof. For a nonzero number of positive valuation b, a prime-exponent power can only have exponent dividing b. For exponent 2 its unit must be 1 modulo 3; for exponent 3 its unit must be ± 1 modulo 9. Thus the three conditions exclude every possible prime exponent. Since $T_{k+9} - T_k = 9k + 45$, the cut-dependent term of the cubic congruence in Theorem 7 depends only on $k \pmod 9$. That theorem identifies $10^{-D(k)}R_{m,k}$ with one of the nine residues to the required precision. Multiplication by $10^{D(k)} \equiv 1 \pmod 9$ preserves valuation and both unit tests. □

Corollary 5 (133 sufficient exclusion classes when $v_3(d) = 2$)**.** *Suppose $v_3(d) = 2$ and put $u = d/9$. Every rotation is excluded from prime-exponent powers over $\mathbb{Q}_3$ if*

$$m \bmod 9 \in \{2,3,5,6\}, \qquad \text{or} \qquad m \bmod 27 \in \{9,17\}, \qquad \text{or} \qquad m \bmod 243 \in E_u,$$

where

$$E_u = \begin{cases} \{26, 27, 81, 108, 134, 135, 216\}, & u \equiv 1 \pmod 3, \\ \{27, 81, 107, 108, 135, 215, 216\}, & u \equiv 2 \pmod 3. \end{cases}$$

The displayed sufficient set consists of exactly 133 classes modulo 243. The converse is established in Theorem 8 below.

Proof. Reduce the cubic congruence modulo 243. Since $d = 9u$, its residue is

$$F_u(m, k) = T_m + \frac{27u}{2}(m^3 - m + 9) - 81umT_k \pmod{243}.$$

It depends only on $m \pmod{243}$, $u \pmod 3$, and $k \pmod 9$: for the dependence on u, use $3 \mid (m^3 - m)$. The first two sets have valuation one, or valuation two with unit 2 modulo 3, just as in Corollary 4. For the additional residues $m = 27, 108, 135, 216 \pmod{243}$, respectively, the normalized unit $F_u(m, k)/27$ is $5, 2, 7, 4 \pmod 9$, independently of k and u. For $m = 81 \pmod{243}$ the valuation is four and the unit is 2 modulo 3. For $u \equiv 1 \pmod 3$, the residues $m = 26, 134 \pmod{243}$ have valuation three and respective normalized-unit sets $\{4, 7\}, \{2, 5\} \pmod 9$ as k varies. For $u \equiv 2 \pmod 3$, $m = 107, 215 \pmod{243}$ give the same two sets. These computations follow by substituting $T_k \pmod 3 \in \{0, 1\}$ in the displayed expression. None of the valuation-three units is ± 1 modulo 9. Valuations one through four have no prime divisor other than 2 or 3, so no further prime exponent is possible. The three displayed sets are disjoint and have sizes 108, 18, and 7, whose sum is 133. □

Theorem 8 (Complete local classification at $v_3(d) = 2$)**.** *Suppose m has d decimal digits, $v_3(d) = 2$, and $u = d/9$. All whole-block rotations fail to be prime-exponent powers in $\mathbb{Q}_3$ if and only if m belongs to the 133 classes in Corollary 5. In every other class, some*

$$k \in \{0, 1, 2, 3, 4, 19, 22\}, \qquad e \in \{2, 3, 5, 7\}$$

satisfy $R_{m,k} \in (\mathbb{Q}_3^\times)^e$.

Proof. Write $X_k = 10^{-D(k)} W_k$, $\Delta = 10^{D(m)} - 1$, and $Y_k = 10^{-D(k)} R_{m,k} = W_m - \Delta X_k$. Multiplication by $10^{D(k)}$ preserves all four power tests: its unit is 1 modulo 9, and raising to the fifth or seventh power is an automorphism of $\mathbb{Z}_3^\times$.

Here is a useful elementary observation. Any full coset of $3\mathbb{Z}/9\mathbb{Z}$ contains either a unit 1 modulo 3, the unit 8 modulo 9, or a residue 3 or 6 modulo 9. Thus, if three values of $Y_k/3^6$ form such a coset, one of the corresponding rotations is a square, cube, or seventh power.

The sufficient direction is Corollary 5. For the converse, substitute the 243 possibilities for m in its polynomial $F_u(m, k)$, at $k = 0, 1, 2, 3$. Apart from the 133 excluded classes, every residue gives a square, cube, or unit (hence fifth power), certified modulo 243, except

$u \pmod 3$	uncertified $m \pmod{243}$
1	$0, 161$
2	$0, 242.$

This finite table uses precisely the tests $b = 0$, or $2 \mid b$ and $w \equiv 1 \pmod 3$, or $3 \mid b$ and $w \equiv \pm 1 \pmod 9$, where $b = v_3(F_u(m, k)) < 5$ and $w = F_u(m, k)/3^b$; zero residues are left uncertified. The companion `verify_h2_local_equivalence.py` checks every entry of this finite table.

Put $a = (10^d - 1)/(9d) - 1$. The binomial expansion shows $v_3(a) = 2$: its leading term is $9(d - 1)/2$, and all subsequent terms have valuation at least three, since the term indexed by $r \geq 3$ has valuation at least $2(r - 1) - v_3(r) \geq 3$. The length identity gives $D(m) = d(m - a)$. If $m \equiv 0 \pmod{243}$, then $v_3(D(m)) = 4$, so $v_3(\Delta) = 6$. The cubic congruence also gives $v_3(W_m) \geq 5$. If this valuation equals five, use $k = 0$. Otherwise write $z = W_m/3^6$ and $c = \Delta/3^6 \in \mathbb{Z}_3^\times$. Since $X_0, X_2, X_3 \equiv 0, 3, 6 \pmod 9$, the residues $Y_k/3^6 = z - cX_k$ at these three cuts form a full coset modulo three, and the observation applies.

The other uncertified classes have $m \equiv -1 \pmod{27}$ and $81 \mid W_m$. Here $v_3(D(m)) = 2$, hence $v_3(\Delta) = 4$. Write $z = W_m/81$ and $c = \Delta/81$, a unit. If $z \equiv 1 \pmod 3$, cut zero is a square. If $z \equiv 0 \pmod 3$, the residues $z - cX_k$ at $k = 0, 2, 3$ include 3 or 6 modulo nine, giving a fifth power. If $z \equiv 2 \pmod 3$ and $c \equiv 1 \pmod 3$, cut one is a square. The remaining case is $z \equiv c \equiv 2 \pmod 3$. Then Y_1 is divisible by 3^5; if its valuation is five, it is a fifth power. Otherwise $3^6 \mid Y_1$, and direct finite arithmetic gives

$$\frac{X_4 - X_1}{9} \equiv 8, \qquad \frac{X_{19} - X_1}{9} \equiv 2, \qquad \frac{X_{22} - X_1}{9} \equiv 5 \pmod 9.$$

Consequently $Y_k/3^6 = Y_1/3^6 - c(X_k - X_1)/9$ at $k = 4, 19, 22$ forms a full coset modulo three. Apply the observation again. All cuts exist, since $d \geq 9$. □

Lemma 3 (One further digit of the prefix congruence)**.** *If $h = v_3(d) \geq 2$ and m has d digits, then*

$$W_m \equiv T_m + \frac{d}{4}\{135 - 21(m^3 - m)\} \pmod{3^{h+5}}.$$

Proof. First, $V_d \equiv 18d \pmod{3^{h+5}}$. Indeed the coefficient bounds and averaging lemma already proved show that all terms of degree at least four in $d^{-1}\sum_{j=1}^{d} f(j)$ vanish modulo 27. The degree-three term also vanishes, since $d^{-1}\sum j^3 = d(d+1)^2/4$ and $v_3(c_3) \geq 3$. The explicit first three coefficients give

$$c_0 \equiv 5, \quad c_1 \equiv 12, \quad c_2/3 \equiv 9 \pmod{27}.$$

Using $9 \mid d$, their averaged contribution is $c_0 + c_1/2 + c_2/6 \equiv 2 \pmod{27}$. The factor $10^{S(d)}$ is 1 modulo 27, proving the boundary assertion. For the tail expansion put $\delta = 10^d - 1$ and $t = \delta - m$. The previously proved expansion, valid modulo 3^{2h+4}, gives

$$W_m \equiv V_d + \frac{t(t-1)}{2} - \delta\frac{t(t^2+5)}{6} \equiv 18d + T_m + \delta\frac{m^3 - m - 3}{6} \pmod{3^{h+5}}.$$

Terms discarded in the substitution have valuation at least $2h + 3 \geq h + 5$. Finally $\delta \equiv 9d - 81d/2 = -63d/2 \pmod{3^{h+5}}$; the multiplying quotient $(m^3 - m - 3)/6$ is 3-adically integral. Substitution gives the result. □

Theorem 9 (Complete local classification at $v_3(d) = 3$)**.** *Let m have d decimal digits, $v_3(d) = 3$, and $u = d/27$. Every whole-block rotation fails to be a prime-exponent power over $\mathbb{Q}_3$ if and only if at least one of the following holds:*

$$m \bmod 9 \in \{2, 3, 5, 6\}, \qquad m \bmod 27 \in \{9, 17\},$$
$$m \bmod 243 \in \{26, 27, 81, 107, 108, 134, 135, 161, 215, 216\},$$
$$m \equiv -729(u + 2) \pmod{6561}.$$

These are exactly 3673 residue classes modulo 6561. Outside them, some $k \in \{0, 1\}$ and $e \in \{2, 3, 5, 7\}$ give $R_{m,k} \in (\mathbb{Q}_3^{\times})^e$.

Proof. Put $X_k = 10^{-D(k)}W_k$, $\Delta = 10^{D(m)} - 1$, and $Y_k = W_m - \Delta X_k$. As before, multiplication by $10^{D(k)}$ preserves all four power tests. Write $a = (10^d - 1)/(9d) - 1$. Then $v_3(a) = 2$, $a/9 \equiv 1 \pmod 3$, and $D(m) = d(m - a)$. Thus $v_3(\Delta) \geq 5$ for all m. The strengthened prefix congruence gives $W_m \equiv T_m \pmod{243}$. If $v_3(W_m) < 5$, every cut has exactly the same valuation and the same square and cube unit tests. Evaluating those tests on $T_m \pmod{243}$ gives the first three displayed sets: $108 + 18 + 10 = 136$ disjoint classes. All other nonzero residues modulo 243 are squares, cubes, or units (the latter are fifth powers).

It remains to treat $243 \mid W_m$. The triangular congruence forces $m \equiv 0, -1 \pmod{243}$. If $v_3(W_m) = 5$, cut zero is a fifth power. For $m \equiv -1 \pmod{243}$, one has $v_3(\Delta) = 5$; therefore, if $v_3(W_m) \geq 6$, cut one has valuation five and is a fifth power. For $m \equiv 0 \pmod{243}$, one has $v_3(\Delta) = 7$ and $\Delta/3^7 \equiv -u \pmod 3$. If $v_3(W_m) \geq 7$, at least one of Y_0, Y_1 has valuation exactly seven, and is a seventh power.

The only remaining case is $m \equiv 0 \pmod{243}$ and $v_3(W_m) = 6$. Write $z = W_m/729$, a unit. Every cut has valuation six, and

$$Y_k/729 \equiv z + 3uT_k \pmod 9.$$

The triangular residues modulo three are exactly $0, 1$, both attained at cuts zero and one. If $z \equiv 1 \pmod 3$, cut zero is a square. For $z \equiv 2 \pmod 3$, squares are impossible and a cube occurs exactly when one of $z, z + 3u$ equals 8 modulo nine. Failure of both conditions is exactly $z \equiv 3u - 1 \pmod 9$. Finally, for $243 \mid m$, the strengthened polynomial reduces to

$$W_m \equiv \frac{2m + 3645u}{4} \pmod{6561}.$$

The exceptional unit condition is therefore equivalent to $m \equiv -729(u + 2) \pmod{6561}$. This extra residue has $m \equiv 0 \pmod{243}$ and is disjoint from the 136 previous classes, giving $136 \cdot 27 + 1 = 3673$. All cuts used exist since $d \geq 27$. □

3 Lifting at boundary primes

Lemma 4 (Boundary periods)**.** *Let $q > 3$ be prime, $r = \operatorname{ord}_q(10)$, and $\gcd(r, 10) = 1$. Choose P with $r \mid P$ and $10^P \equiv 1 \pmod{9r}$. Then $S(d) \pmod r$ has period P, and $V_d \pmod q$ has period qP. More precisely, for $t \geq 0$ and $0 \leq s < P$,*

$$V_{tP+s} \equiv tV_P 10^{S(s)} + V_s \pmod q. \tag{15}$$

Proof. In a complete j-digit block let $a = 10^{j-1}$, $B = 10^j$, and $N = 9a$. Appending that block acts by $z \mapsto B^N z + U(j)$. Modulo q, if $B \neq 1$,

$$U(j) = \frac{(a(B-1)+1)B^N - (B(B-1)+1)}{(B-1)^2}; \tag{16}$$

if $B = 1$, it is $Na + N(N-1)/2$. The affine map therefore depends only on $a, B, N \pmod q$ and $N \pmod r$. These data repeat after P steps: $r \mid P$ fixes the first three, and $10^P \equiv 1 \pmod r$ fixes the last. Also $9j10^{j-1} \pmod r$ has period P, and

$$S(P) = \frac{(9P-1)10^P + 1}{9} \equiv 0 \pmod r$$

by the stronger modulus $9r$. Thus $S(d) \pmod r$ is periodic, and a complete P-cycle acts by $z \mapsto z + V_P$ modulo q. Appending s more blocks proves (15), and adding q cycles proves the asserted period of V_d. □

Proof of Theorem 4. We interpolate a complete P-cycle of digit blocks and average it over an arbitrary number of cycles, retaining division by that number. Put

$$\lambda = \frac{\log(10^r)}{r} \in p\mathbb{Z}_p, \qquad \omega = 10e^{-\lambda}, \qquad L = P\lambda.$$

Then $\omega^r = 1$, and its reduction has order r. For $s = 1, \ldots, P$ and $X \in \mathbb{Z}_p$, set

$$j = PX + s, \quad B = 10^s e^{LX}, \quad A = B/10, \quad N = 9A, \quad b = B - 1, \quad S_s(X) = \frac{(9j-1)B + 1}{9}.$$

Let $E_s = 9s10^{s-1}$ and $H_s = S(s)$ be fixed integers. Define

$$U_s(X) = \frac{(Ab+1)\omega^{E_s} e^{\lambda jN} - (Bb+1)}{b^2}, \qquad F(X) = \sum_{s=1}^{P} U_s(X)\omega^{-H_s} e^{-\lambda S_s(X)}.$$

For integral $X \geq 0$, the period conditions give $jN \equiv E_s \pmod r$ and $S(j) \equiv H_s \pmod r$. Thus $U_s(X)$ is the actual j-digit block and $F(X) = \sum_{j=PX+1}^{P(X+1)} U_j 10^{-S(j)}$. For $r \mid s$, remove the apparent singularity by

$$U_s(X) = AN + (Ab+1)\sum_{t \geq 0} \binom{N}{t+2} b^t. \tag{17}$$

Here $b \in p\mathbb{Z}_p$ throughout $\mathbb{Z}_p$; if $r \nmid s$, b is a unit.

Write $F(X) = \sum_{n \geq 0} c_n X^n$. The coefficient bound is

$$v_p(c_n) \geq n - v_p(n!) - \left\lfloor \frac{n+1}{p-1} \right\rfloor \geq \left\lceil \frac{p-3}{p-1} n \right\rceil \quad (n \geq 1). \tag{18}$$

To see this, each nth derivative of e^{LX} gains at least p^n. The same holds for the nth derivatives of λjN and $-\lambda S_s$, for $n \geq 1$: a derivative of the linear factor j loses at most one L, compensated by λ. Exponentiating preserves the property, as do inverses of unit denominators. In the tth summand of (17), an undifferentiated factor b contributes p, while a factor differentiated $i \geq 1$ times contributes p^i. Combining all factors gives at least $p^{\max(t,n)}$ for the nth derivative, before the denominator $(t+2)!$. It loses at most $\lfloor (t+1)/(p-1) \rfloor$ powers of p. For $t \leq n$ this gives the asserted bound, and for $t > n$ use that $t - \lfloor (t+1)/(p-1) \rfloor$ is nondecreasing. The tails converge with all derivatives. The normalizing exponential preserves the bound; division by $n!$ gives the first inequality. The second uses $v_p(n!) \leq (n-1)/(p-1)$.

For $M \geq 1$, the exact Stirling identity

$$\frac{1}{M}\sum_{i=0}^{M-1} i^n = \sum_{t=0}^{n} \left\{ {n \atop t} \right\} \frac{t!}{t+1} \binom{M-1}{t}$$

bounds its valuation below by $-\lfloor \log_p(n+1) \rfloor$. For $p \geq 7$, (18) exceeds this loss by at least one: if $k = \lfloor \log_p(n+1) \rfloor \geq 1$, then $n \geq p^k - 1$ makes its final bound at least $(p-3)k \geq k+1$; the case $k = 0$ is immediate. Termwise averaging therefore gives

$$\frac{1}{M} \sum_{i=0}^{M-1} F(i) \equiv c_0 \equiv C \pmod{p}.$$

The last equality follows by concatenating the first P blocks and using $r \mid S(P)$. Likewise $V_{PM} = 10^{S(PM)} \sum_{i=0}^{M-1} F(i)$ and $r \mid S(PM)$ give the exact valuation $v_p(M)$ and normalized unit $CM/p^{v_p(M)}$.

Finally put $d = PM$ and $\nu = v_p(10^r - 1) \geq 1$. Lifting the exponent gives $v_p(10^d - 1) = \nu + a$. In $S(d) = d10^d - (10^d - 1)/9$, the summands have unequal valuations a and $\nu + a$, so $v_p(S(d)) = a$. Since $r \mid S(d)$,

$$v_p(10^{S(d)} - 1) = \nu + a \geq a + 1.$$

The rotation identity (1) now proves both assertions in (5), including the factor contributed by the cut. □

Proof of Theorem 5. Before using $C \neq 0$, the preceding averaging argument gives

$$V_d \equiv C \frac{d}{P} \pmod{p^{a+1}}.$$

Put $B = 10^d$ and $t = B - 1 - m$. The removed t integer blocks all have length d, so the exact tail identity is

$$V_d = B^t W_m + \sum_{j=1}^{t} (B - j) B^{j-1}.$$

Lifting the exponent gives $B \equiv 1 \pmod{p^{a+1}}$. Thus the tail sum is $-t(t-1)/2$, and $t \equiv -m$ gives $W_m \equiv V_d + m(m+1)/2$ at this precision. Also $D(m) = S(d) - dt$ is divisible by both r and p^a; the latter follows from $S(d) = d10^d - (10^d - 1)/9$. Hence $10^{D(m)} \equiv 1 \pmod{p^{a+1}}$, and (1) proves (6).

When $p \mid d/P$, the polynomial in parentheses reduces modulo p to $m(m+1)/2$. Its roots 0 and -1 are simple. Each has p lifts modulo p^2, exactly one of which is zero modulo p^2. The other $p-1$ lifts have valuation one, proving the count. At $p = 37$, $P = 3$, $d = 111u$, the polynomial is $m(m+1)/2 + 1221u \pmod{1369}$; its two zero lifts are $296u$ and $1073u - 1$.

A further whole-block exclusion is $d \equiv 3 \pmod{111}$, $m \equiv 126 \pmod{999}$ within the d-digit interval. The elementary identities $W_m \equiv m(m+1)/2 + 9d\lfloor (m+1)/3 \rfloor \pmod{27}$ and $D(m) \equiv dm \pmod{9}$ (proved in `RESIDUE_EXCLUSIONS.md`) give $R_{m,k} \equiv 9 \pmod{27}$, forcing the exponent to be 2. But (6) has polynomial residue 5 modulo 37, a nonsquare, and 10 is a square modulo 37. This excludes every rotation. □

Proof of Corollary 2. For a finite set $\mathcal{S}$ of primes $p \geq 7$ with $\gcd(\mathrm{ord}_p(10), 10) = 1$, choose $P_p = \mathrm{lcm}(r_p, \mathrm{ord}_{9r_p}(10))$. This satisfies the period hypotheses and $p \nmid P_p$: every prime factor of $r_p \varphi(9r_p)$ is smaller than p. Set

$$L = \mathrm{lcm}_{p \in \mathcal{S}}(P_p p), \quad M = \prod_{p \in \mathcal{S}} p^2, \quad \theta = \prod_{p \in \mathcal{S}} \left(1 - \frac{2(p-1)}{p^2}\right).$$

For fixed $L \mid d$, the Chinese remainder theorem shows that the fraction of classes modulo M avoiding every valuation-one obstruction is exactly θ. Thus, with $H = 9 \cdot 10^{d-1}$, at least $H(1-\theta) - M$ values of m have a valuation-one prime shared by every rotation, so none of their rotations is powerful.

Primes $p \equiv 3 \pmod{40}$ supply a divergent reciprocal sum by Dirichlet's theorem [4]. Quadratic reciprocity gives $(10/p) = 1$, and $(p-1)/2 \equiv 1 \pmod{10}$; therefore r_p is odd and coprime to 10. Consequently θ can be made smaller than $\varepsilon/2$. Choose such a finite set and then d_0 so that $M/H < \varepsilon/2$ for $d \geq d_0$. □

Proof of Theorem 3. For both pairs in Theorem 3,

$$\mathrm{ord}_p(10) = s \mid 9, \qquad v_p(10^s - 1) = 1, \qquad S(j) := D(10^j - 1) \in 9\mathbb{Z}.$$

Let U_j concatenate the j-digit integers, and put $B = 10^j$, $b = B - 1$, $a = B/10$, $q = 9a$. Its length jq is divisible by s. If $s \nmid j$, the block formula used in (11) gives $U_j \equiv -9a/b \pmod{p}$. If $s \mid j$, its multiplier B is one modulo p, so

summing the appended integers gives $U_j \equiv 9/200 \pmod p$. Thus $U_j \pmod p$ has period s. Pairing each nonidentity sth root $z = 10^j$ with z^{-1} yields

$$\sum_{j=1}^{s-1} \frac{10^j}{10^j - 1} = \frac{s-1}{2}, \qquad \sum_{j=1}^{s} U_j \equiv -\frac{9(s-1)}{20} + \frac{9}{200} = C_s \pmod p.$$

All block multipliers are one modulo p, proving (4), including its rotation assertion. This proves Theorem 3. Its corresponding valuations and normalized units follow from Theorem 4 with $P = s$.

□

4 Power characters and periodic transfer

For a prime q and a prime $e \mid q-1$, put $\chi_{q,e}(z) = z^{(q-1)/e} \pmod q$. A nonzero eth power has character 1. If

$$10^{S(d)} \equiv 1 \pmod q, \qquad \chi_{q,e}(10) = 1, \qquad \chi_{q,e}(V_d) \notin \{0, 1\}, \tag{19}$$

then (1) gives the same nontrivial character for every $R_{10^d-1,k}$. Thus none is an eth power. When $v_3(d) + 2$ is a power of e, Theorem 2 eliminates all other prime exponents.

Here are the finite inputs. Their nonzero cycle constants also verify every hypothesis of Theorem 4:

i	q_i	r_i	P_i	$V_{P_i} \pmod{q_i}$
1	37	3	3	33
2	757	27	27	538
3	333667	9	9	221885
4	440334654777631	27	27	178343635184937
5	1527791	43	903	547718
6	43037	29	812	409

For each row of the next table, $r_i \mid S(d)$, $\chi_{q_i,e}(10) = 1$, and the displayed character is nonzero and different from 1. Each $v_3(d) + 2$ is a power of the listed e.

d	e	i	$\chi_{q_i,e}(V_d)$
$2, 14, 20$	2	2	-1
$5, 26$	2	3	-1
$8, 11, 17, 18$	2	1	-1
$12, 15$	3	1	26
23	2	4	-1
27	5	5	990998
54	5	4	386179030766766
243	7	6	12819

The accompanying `BOUNDARY_CHARACTERS.json` records these exact inputs; `verify_boundary_characters.py` verifies them. Their finite arithmetic consists of complete factorizations of $q_i - 1$, Lucas primality witnesses, modular powers, and the recurrence $V_j = B^N V_{j-1} + U(j)$ with (16). An independent verification uses exact matrix powering for every listed d. For a Lucas witness a, the checked conditions are $a^{q-1} \equiv 1 \pmod q$ and $\gcd(a^{(q-1)/\ell} - 1, q) = 1$ for every prime $\ell \mid q-1$; the factor primes are verified by trial division. These conditions force every prime divisor of q to be at least q, proving primality.

Proof of Theorem 6. Corollary 1 and the table exhaust $d = 1, \ldots, 27$; the last two rows handle 54 and 243. Equation (19) excludes every available prime exponent.

For any row put $h = v_3(d)$ and $T = q_i \operatorname{lcm}(P_i, 3^{h+1})$. Replacing d by $d + tT$ preserves h, $S(d) \pmod{r_i}$, and $V_d \pmod{q_i}$, by Lemma 4. Hence the same character argument works for all $t \geq 0$. The rows $d = 12, 15, 27, 243$ give exactly (7). □

Corollary 6 (Exact local power criteria)**.** *Put $Q = 440334654777631$ and let u be a positive integer. For every rotation at the indicated boundary,*

d	*equivalence over the stated field*
$27u$	*R is a fifth power in $\mathbb{Q}_Q$ $\iff$ u is a fifth power in $\mathbb{Q}_Q$*
$243u$	*R is a seventh power in $\mathbb{Q}_{757}$ $\iff$ u is a seventh power in $\mathbb{Q}_{757}$.*

Explicitly, for $(p,e)=(Q,5)$ or $(757,7)$, write $u=p^a t$ with $p\nmid t$. The local condition is exactly

$$e \mid a, \qquad t^{(p-1)/e} \equiv 1 \pmod{p}. \tag{20}$$

If $3\nmid u$, failure of this condition excludes every integer perfect power in the whole rotation block.

Proof. Theorem 4 gives $v_p(R)=v_p(u)=a$ and

$$R/p^a \equiv 10^{D(k)}178343635184937\,t \pmod{Q}, \qquad R/p^a \equiv 10^{D(k)}300t \pmod{757},$$

respectively. The displayed coefficients and 10 have character 1 for exponents 5 and 7 respectively, as checked in the certificates. Since $e\neq p$, Hensel's lemma lifts every nonzero eth root modulo p. Thus a number in $\mathbb{Q}_p$ is an eth power exactly when its valuation is divisible by e and its normalized unit is an eth power modulo p. This proves both equivalences. When $3\nmid u$, the only possible prime exponent is 5 or 7 respectively, by Theorem 2. □

Proof of Corollary 3. The 3-adic restrictions leave only the following disjoint sets:

$$d\equiv 2 \pmod 3, \qquad d\equiv 12,15 \pmod{27}, \qquad d\equiv 18 \pmod{27}, \qquad 27\mid d.$$

In the first and third sets, a perfect power must be a square; in the second, a cube. Put $Q=440334654777631$. In every fixed class modulo 27, Lemma 4 makes V_d affine modulo each of $37, 333667, 757, Q$. In the class $d\equiv a \pmod{27}$ the respective slopes are $1, 331988, 538, 178343635184937$ multiplied by $10^{S(a)}$; all are nonzero. The four residues are independent by the Chinese remainder theorem. For a square candidate the exact valuation of $S(d)$ from (12) shows $27\mid S(d)$. Thus all four rotation characters are fixed, and the proportion of residues that are zero or squares at all four primes is

$$\alpha=\frac{19}{37}\frac{166834}{333667}\frac{379}{757}\frac{Q+1}{2Q}.$$

For cubes use 37 and 333667, giving the proportion $\beta=(13/37)(111223/333667)$, as in (4). The upper density of the remaining d is therefore at most $(10\alpha+2\beta+1)/27$.

For a prime $e\mid p-1$, the density of positive integers that are eth powers in $\mathbb{Q}_p$ is

$$s(p,e)=\sum_{j\ge 0}\frac{p-1}{ep^{ej+1}}=\frac{(p-1)p^{e-1}}{e(p^e-1)}.$$

Indeed, the valuation must be a multiple of e, and exactly $(p-1)/e$ normalized units modulo p are eth powers. Truncating at valuation K leaves a subset of multiples of p^K, of density p^{-K}, proving the density assertion. For $p\neq 3$, the same proportion applies conditional on $3\nmid u$, by the Chinese remainder theorem and the same truncation. Thus, inside $27\mid d$, Corollary 6 excludes disjoint sets of densities

$$\delta_5=\frac{2}{81}\big(1-s(Q,5)\big), \qquad \delta_7=\frac{2}{729}\big(1-s(757,7)\big).$$

Here $v_3(d)$ is respectively 3 or 5; positive local valuations are included. Also $d=81u$ with $u\equiv 1,7 \pmod 9$ is excluded by Corollary 1, adding the disjoint density $2/729$. For $h=v_3(d)\in\{6,7,10,14,16\}$, Corollary 1 adds disjoint exclusions of total density

$$E=\frac{1}{2187}+\frac{4}{19683}+\frac{2}{531441}+\frac{1}{14348907}+\frac{2}{387420489}.$$

These follow respectively from $h+2=8,9,12,16,18$ and the indicated unit classes in that corollary. None overlaps the already counted sets, which have $h\le 5$. Consequently the exact rational lower bound is

$$\rho_*:=1-\frac{10\alpha+2\beta+1}{27}+\delta_5+\delta_7+\frac{2}{729}+E>0.95599527150.$$

The companion `verify_density_refinement.py` checks this inequality using rational arithmetic. □

5 A uniform sieve for powerful values

We now prove Theorem 1. In a fixed d-digit interval set

$$B = g^d, \quad L = g^{d-1}, \quad c = B - 1, \quad A = c^2 W_{L-1} + Lc + 1.$$

The recurrence $W_{m+1} = BW_m + (m+1)$ gives, for $L - 1 \le m \le B - 1$,

$$c^2 W_m = AB^{m-L+1} - ((m+1)c + 1), \qquad D(m) = D(L-1) + d(m - L + 1).$$

For fixed k, writing $u = g^{D(k)}$, substitution into (1) gives

$$R_{m,k} = \alpha B^m + \beta m + \gamma, \quad \begin{cases} \alpha = (uA/c^2 - g^{D(L-1)} W_k) B^{1-L}, \\ \beta = -u/c, \quad \gamma = W_k - uB/c^2. \end{cases} \tag{21}$$

At every prime $p \nmid g(B-1)$ the coefficients are defined modulo p^2 and β is a unit. The right side provides a periodic modular extension; only its values inside the digit interval are identified with the concatenation.

Lemma 5 (Valuation-one densities)**.** *Let $p \nmid B\beta$ and $F(n) = \alpha B^n + \beta n + \gamma \pmod{p^2}$. The event $p \mid F(n)$, $p^2 \nmid F(n)$ has period dividing $T_p = p^2(p-1)$ and density $\rho_p = (p-1)/p^2$. For any finite set P of eligible primes the events have joint density $\prod_{p \in P} \rho_p$ in a period dividing $\operatorname{lcm}_{p \in P} T_p$.*

Proof. Fix $n \pmod{p-1}$. The exponential term is constant modulo p, and the linear coefficient is nonzero, so exactly one class $n \pmod p$ is a root. Its p lifts differ by $p(p-1)t$. Euler's theorem fixes the exponential term modulo p^2 on these lifts, while the linear term takes p distinct values. Precisely one lift is zero modulo p^2, leaving $p - 1$ of valuation one.

For the joint assertion induct on the largest prime q. The smaller-prime events have a period T with no prime divisor as large as q. Thus $q \nmid H = \operatorname{lcm}(T, q-1)$. In each fixed class modulo H, the smaller-prime events are fixed and the Chinese remainder theorem lets $n \pmod{q^2}$ vary freely. The local count gives the same fraction ρ_q in every class. Averaging multiplies the previous joint density by ρ_q. This allows overlapping periods, including a smaller prime dividing a larger prime minus one. □

For an interval I of H integers and a subset P', Lemma 5 gives its simultaneous event count as $H \prod_{p \in P'} \rho_p + O(\prod_{p \in P'} p^3)$, since the interval error is at most the joint period. Take eligible primes at most z, put $\lambda = \sum \rho_p$, and choose an even integer $K \ge 0$. Powerful values avoid every valuation-one event. Finite Bonferroni gives

$$\#\{n \in I : F(n) \text{ powerful}\} \le H \left(e^{-\lambda} + \frac{\lambda^{K+1}}{(K+1)!} \right) + (K+1) z^{4K}. \tag{22}$$

Indeed, if e_j are the elementary symmetric sums of the ρ_p, the even truncated main term is at most $\prod(1-\rho_p) + e_{K+1} \le e^{-\lambda} + \lambda^{K+1}/(K+1)!$. There are at most z^j subsets of size j, each costing at most z^{3j}, which proves the displayed error. These estimates are uniform in the coefficients of (21) and in k.

Choose

$$\eta = \frac{\log g}{100}, \qquad z = \exp\left(\frac{\eta d}{\log d} \right), \qquad K = 2\lceil 4 \log d \rceil,$$

and use primes $d^2 < p \le z$ with $p \nmid g(g^d - 1)$. For large d, $z > d^2$. Omitted primes exceeding d^2 and dividing $g^d - 1$ number at most $\log(g^d - 1)/(2 \log d)$, with reciprocal sum $O_g(1/(d \log d))$. The fixed prime divisors of g eventually lie below d^2. Mertens' estimate [3] therefore yields

$$\lambda = \log\log z - \log\log(d^2) + O_g(1) = \log d - 2 \log\log d + O_g(1).$$

In particular $0 < \lambda \le \log d$ eventually, so $e^{-\lambda} \ll_g (\log d)^2/d$. Also

$$\frac{\lambda^{K+1}}{(K+1)!} \le (e/8)^{K+1} \le d^{-8}, \qquad (K+1) z^{4K} < g^{d/2}$$

for large d; the latter follows from $K \le 10 \log d$ and $\log((K+1)z^{4K}) \le \log(K+1) + 40\eta d < d \log g/2$. Thus for every fixed k and every interval inside the digit block with $m > k$,

$$\#\{m \in I : R_{m,k} \text{ powerful}\} \ll_g |I| \frac{(\log d)^2}{d} + g^{d/2}. \tag{23}$$

Let D be the digit length of $\lfloor X \rfloor$. For each $d \le D$ and k, apply (23) on $\max(g^{d-1}, k+1, 2) \le m \le \min(X, g^d - 1)$. There are at most g^d choices of k, so the total interval error is $\sum_{d \le D} g^{3d/2} = O_g(X^{3/2})$. Blocks with $d \le D/2$ contain only $O_g(X)$ pairs and may be bounded trivially. On the other blocks the main factor is $O((\log D)^2/D)$, and there are $O(X^2)$ pairs. This proves (2).

Finally, each m supplies m distinct rotations. If m has $d \ge 2$ digits, the cyclic digit word has a unique occurrence of 1 followed by $d-1$ zeros: it is the integer g^{d-1}. Such a zero run cannot cross an integer boundary or the cyclic join, whose next digit is 1. The unique marker prevents a nonidentity digit rotation from fixing the word. For $d = 1$ all digits are distinct. Digit lengths increase strictly with m, so the sequence through block $m = X$ has $X(X+1)/2$ entries. Bounding a partial final block by the whole block and taking $X \asymp \sqrt{N}$ proves the sequence-index assertion. □

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